

Multichannel Haptic Communication Platform with Wearable Sensing and Display

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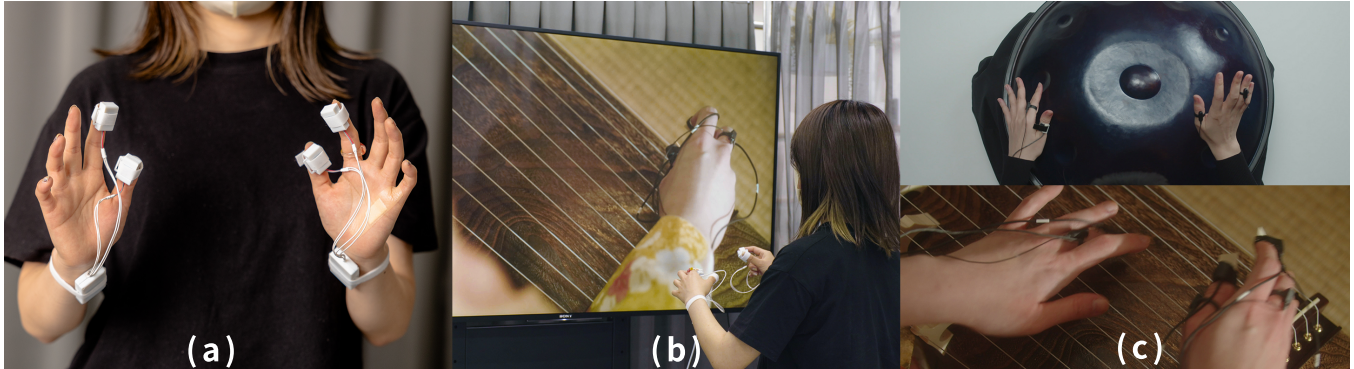


Figure 1: This is a multi-channel vibrotactile recording, editing, and playback platform. (a) Prototype of a 4-channel fingertip-mounted vibrotactile device. (b) A person wearing the device reliving a Koto performance with video. (c) Example of 4-channel haptic contents.

ABSTRACT

Our goal has been to create an environment in which anyone can take advantage of tactile technology. Content that incorporates tactile sensations in addition to visual and auditory sensations will activate people's interaction with it. On the other hand, both sensing and display are complicated to create high-quality content, and ease of use is lost. In this paper, we propose a platform that can easily develop multi-channel vibrotactile contents by utilizing

existing workflows. By preparing a 4-channel fingertip-mounted vibrotactile device as a display, a high-resolution experience with less load during the experience is realized. We demonstrate sensor-based pre-recorded or real-time multi-channel tactile contents and show the concept of near-future tactile technology including use cases.

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1 INTRODUCTION

We have been developing a tactile prototyping tool, the TECHTILE Toolkit [Minamizawa et al. 2012], to create an environment in which anyone can utilize tactile technology. The goal is to realize the recording and reproduction of sensations using not only auditory and visual senses but also tactile senses, and to activate interaction between people in online communication. When considering the transmission of spatially and temporally high-resolution haptic information such as playing a musical instrument, increasing the number of fingertips, or channels, to be touched is effective in improving the quality of the experience. However, both sensing and actuation have their own challenges, such as increased weight and complicated workflow.

We propose a platform for recording, editing, and playback of multi-channel vibrotactile sensations that does not interfere with the movements of the performer and audience. This enables the transmission of the sensations of others with higher resolution while reducing the burden on the human operator during recording and playback. By leveraging existing content development workflows, we promote the democratization of multi-channel tactile content. This paper describes a lightweight, multi-channel tactile display, as well as multi-channel content and its development methodology, up to real-time tactile sharing. We aim to include tactile sharing in the communication infrastructure of the 6G era in the future, and this research is a stepping stone to that goal.

2 DESIGN AND IMPLEMENTATIONS

2.1 System Design

As shown in Figure 2, the production of multi-channel vibrotactile content consists of the processes of recording, editing, and playback, each of which inherently requires specialized skills and experience. Therefore, we propose to utilize existing workflows for the development of video and audio content. By including vibrotactile information in multi-channel audio, multiple tactile data can be synchronized and coexist with audio and video, making it easy to develop multimodal content. This also allows for real-time transmission of multi-channel vibrotactile sensations.

As a device for presenting multi-channel vibrotactile sensation, we have devised a fingertip-mounted 4-channel vibrotactile device shown in Figure 1a. The device operates as a 2-channel USB audio device per hand and can be operated from many devices such as PCs and smartphones. The fabricated device weighs approximately 35g per side and consists of two vibration actuators, a drive circuit, and a 3D printer housing. The driving circuit consists mainly of a microcontroller (Seeeduino Xiao RP2040, Seeed Technology Co., Ltd.) and two digital power amplifiers (NS4168, Shenzhenshi YONGFUKANG Technology co., Ltd.). It operates as a stereo USB Audio device. The vibration actuator is fixed to the fingertip with an adjustable belt, while the drive circuit is designed to be worn on the arm. By fixing the circuit board to the arm, we designed the device so that the weight of the cables and the board itself does not become a burden on the fingers. In this way, the user can freely move the fingertips and imitate physical movements, thereby enhancing the quality of the experience. By wearing devices on both hands and



Figure 2: Development flow of vibrotactile content using existing tools

connecting them to a PC (Mac mini), devices operates as a multi-channel audio interface, and the user experiences multimodal tactile content synchronized with audio and video.

2.2 Content Design

For tactile recording, an accelerometer (2302B, Showasokki Co., Ltd.), a PVDF sensor [Tanaka et al. 2015], and a contact microphone (EM246U100B1, Primo) were attached to a finger or target object, and multiple signals were recorded simultaneously with a field recorder (Portacapture X8, TEAC Corp.). By fixing the sensor with the jig we created, we can acquire sensations without interfering with finger movements or interactions with the object (Figure 1c). These data were edited as multi-channel audio video using Premiere Pro video editing software, and mixed down separately for thumb and other finger vibrations to create 4-channel content. Since vibrations from other fingers are less likely to propagate to the thumb [Shao et al. 2016], the device was worn on two separate channels, one for the thumb and the other for the other fingers, to increase the resolution of the experience while reducing the number of channels.

3 USER EXPERIENCES

At SIGGRAPH Asia 2023, participants can actually wear the device on both hands and experience various sensations together with vibrotactile sensations. Specifically, by presenting the device with pre-recorded vibrotactile sensations along with images of the koto and hand pan, participants can simulate the sensations of a professional performance.

Then, we plan to conduct live demonstrations in which users can experience the sensation of touching an object or playing a musical instrument through real-time transmission of multi-channel tactile sensations. The feasibility of the concept will be demonstrated with an eye toward future use cases.

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